W07 Pre-lab Quiz: Smart Dimmer

Started: Feb 16 at 6:15pm

Quiz Instructions

Week 07

Pre-Lab Quiz

Review all the study materials for the module, then use what you learned and the following information to answer the questions.

Smart Light Dimmer Using an ADC

A smart home light dimmer uses an analog-to-digital converter (ADC) to adjust the brightness of an LED light based on the position of a rotary potentiometer. The potentiometer acts as a voltage divider, producing an analog voltage between 0 V and 3.3 V. This voltage is sampled by a microcontroller's ADC, which then maps the digital value to a pulse-width modulation (PWM) duty cycle to control the LED brightness.

System Specifications:

- **ADC Reference Voltage**: 3.3 V
- **ADC Resolution**: Configurable between 8-bit, 10-bit, and 12-bit modes.
- **Sampling Rate**: 1 kHz
- **PWM Output**: 0% to 100% duty cycle based on ADC reading.

Question 1 0.5 pts

In the smart light dimmer system, the potentiometer is typically adjusted by hand at around 5 Hz. What is the minimum ADC sampling rate required to accurately capture this motion?

- (selected) At least 10 Hz, to satisfy the Nyquist sampling criterion.
- At least 5 MSPS, to match the ADC's maximum capability.
- At least 100 Hz, to ensure high-resolution tracking.
- At least 5 Hz, equal to the potentiometer's frequency.

○
At least 1 kHz, matching the system's default sampling rate.

Question 2 0.5 pts

In the smart light dimmer system, a PWM signal is smoothed by a low-pass filter to control LED brightness. How do the PWM frequency and duty cycle affect the smoothness and brightness of the output?

○
A lower PWM frequency results in a smoother output voltage.

◉
Increasing the PWM frequency makes the output smoother, while the duty cycle determines brightness.

○
Increasing the duty cycle makes the output smoother, while the frequency determines brightness.

○
Both frequency and duty cycle only affect brightness, not smoothness.

○
The duty cycle affects smoothness, and frequency has no effect.

Question 3 0.5 pts

If a user notices that the light brightness changes in noticeable steps rather than smoothly, what is the likely cause?

○
The potentiometer is faulty and producing inconsistent voltages.

○
The LED is not compatible with PWM control.

○
The sampling rate is too fast, overwhelming the system.

○
The PWM frequency is too high, causing instability.

◉
The ADC resolution is too low, resulting in coarse brightness steps.

Question 4 0.5 pts

What is the effect of increasing the ADC resolution from 8-bit to 12-bit in a microcontroller-based system?

○
Eliminate the need for a low-pass filter.

○
Decrease the reference voltage required.

- [ ] Reduce the input voltage range of the ADC.
- [x] Increase the number of discrete digital output levels.
- [ ] Increase the sampling rate of the ADC.

Question 5 0.5 pts

What is the smallest detectable change in input voltage (quantization step size) when the ADC operates in 8-bit mode with a reference voltage of 3.3 V?

0.0129

Question 6 0.5 pts

The smart light dimmer system uses a 10-turn potentiometer with 100 discrete clicks per turn, producing a voltage from 0 V to 3.3 V. This voltage is read by a 12-bit ADC. How does the voltage step size of the potentiometer compare to the ADC's quantization step size?

- [ ] The ADC cannot detect changes from the potentiometer due to mismatched voltage ranges.
- [ ] The ADC's resolution is too high, which causes noise in the readings.
- [ ] The potentiometer's voltage steps are smaller than the ADC's quantization steps, causing some steps to be missed.
- [ ] The potentiometer and ADC have the same resolution, so each click maps to one ADC level.
- [x] The potentiometer's voltage steps are larger than the ADC's quantization steps, allowing the ADC to capture each change accurately.

Question 7 1 pts

Which of the following is a speculation-based lock-free synchronization technique?

- [ ] Mutex locks
- [ ] Memory barriers
- [ ] Spinlocks
- [ ] Load balancing

- [x] Hardware transactional memory (HTM)

Question 8 1 pts

Which instruction is used in Power Architecture to enforce memory ordering?

- [ ] lock
- [ ] strex
- [x] sync
- [ ] cmpxchg
- [ ] mfence

Question 9 1 pts

Why is synchronization scalability a concern in multicore systems?

- [ ] Because caches are disabled.
- [ ] Because memory becomes non-volatile.
- [ ] Because interrupts are disabled.
- [ ] Because instruction sets become incompatible.
- [x] Because contention increases with the number of cores.

Question 10 1 pts

Why is DMA preferred over CPU for large data transfers?

- [ ] DMA is easier to program.
- [ ] DMA uses more memory bandwidth.
- [ ] DMA requires no configuration.

- [x] DMA consumes less power and avoids CPU stalls.
- [ ] DMA is slower but more reliable.

Question 11 1 pts

Which peripheral communication method is most suitable for unpredictable data arrival?

- [x] Interrupt processing
- [ ] Polling
- [ ] Time-based processing
- [ ] Manual triggering
- [ ] Scheduled DMA

Question 12 1 pts

Which method of core-to-peripheral communication is the least power-efficient?

- [ ] DMA signaling
- [ ] Time-based processing
- [ ] Interrupt processing
- [ ] Event-driven scheduling
- [x] Polling

Question 13 1 pts

What is the function of the msgsnd instruction in Power Architecture?

- [x] It sends an inter-processor interrupt message.
- [ ] It flushes the instruction pipeline.

It synchronizes memory accesses.

It clears the message buffer.

It locks a shared memory region.

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